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**DMoEBERT : A Sparse Mixture-of-Experts Encoder
for Efficient Natural Language Inference in
Resource-Constrained Environments**

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A thesis submitted in partial fulfillment of the requirements for the degree of “Bachelor of Science in Computer Science & Engineering”

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Author

Abstract

Large language models have achieved strong performance on natural language inference (NLI), but their high parameter counts and training costs limit deployment in resource-constrained environments. We present DMoEBERT, an encoder-only transformer architecture that integrates Dynamic Tanh with a noisy top-k routed sparsely activated Mixture of Experts (MoE) for efficient NLI. DMoEBERT-23M achieves 82.7% and DMoEBERT-45M achieves 83.7% accuracy on Stanford NLI dataset with only 2 hr 37 min and 3 hr 43 min training time respectively, using a single P100 GPU. Through comprehensive ablation studies against fine-tuned BERT-base and RoBERTa-base, we validate the effectiveness of our architectural design choices, particularly the integration of Dynamic Tanh and noisy top-K expert selection in MoE layers in encoder-only models. Compared to these baselines, DMoEBERT-23M requires 13.5 times and DMoEBERT-45M requires 6 times fewer FLOPs while delivering competitive accuracy, making it well-suited for practical deployment in resource-constrained settings.

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CHAPTER I

Introduction

1.1 Background

Natural Language Inference (NLI) is a primary task in natural language processing that requires models to determine the logical relationship between a premise and hypothesis, whether the hypothesis is entailed by, contradicts, or is neutral with respect to the premise. This task serves as a crucial benchmark for evaluating machine reading comprehension and logical reasoning capabilities, with applications directly related to question answering, dialogue systems, and text summarization.

The introduction of large-scale datasets such as the Stanford Natural Language Inference (SNLI) corpus [1], Multi-Genre Natural Language Inference (MultiNLI) [2], and Adversarial Natural Language Inference (ANLI) [3] have enabled the development and evaluation of increasingly sophisticated models, culminating in transformer-based architectures that have achieved remarkable performance gains.

The transformer architecture [10], particularly through models like BERT [8] and RoBERTa [9], has revolutionized NLI performance by leveraging bidirectional attention mechanisms and large-scale pre-training. However, these models typically contain hundreds of millions of parameters (BERT-base: 110M, RoBERTa-base: 125M), requiring substantial computational resources for both training and inference.

Recent research has explored two promising directions for improving efficiency: Mixture-of-Experts (MoE) architectures [4, 5] that achieve sparse activation through expert routing, and normalization-free training approaches such as Dynamic Tanh [7] that eliminate the computational overhead of normalization layers. While MoE has demonstrated success in large-scale language models like Switch Transformer [6] and GLaM [13], and Dynamic Tanh has shown promise in various architectures, their integration in resource-constrained settings remains underexplored.

1.2 Problem Statement

Despite the remarkable success of large pre-trained models in NLI tasks, several critical challenges limit their accessibility and practical deployment:

Computational Barriers: State-of-the-art NLI models require extensive computational resources for training and inference. BERT-base [8] and RoBERTa-base [9] models, while

achieving high accuracy, demand significant GPU memory and training time, making them inaccessible to researchers and practitioners with limited resources.

Parameter Efficiency Limitations: Although parameter-efficient fine-tuning methods such as Low-Rank Adaptation (LoRA) [11] reduce the number of trainable parameters, they still rely on large pre-trained base models, inheriting the associated memory and computational requirements during inference.

Scalability Gap: While MoE architectures have demonstrated efficiency gains in trillion-parameter models [6, 14], their effectiveness in smaller, resource-constrained settings remains unclear. Similarly, normalization-free approaches like Dynamic Tanh [7] have not been thoroughly evaluated in conjunction with sparse expert routing mechanisms.

Deployment Constraints: Edge computing scenarios, mobile applications, and resource-limited environments require models that can deliver competitive performance while operating within strict memory, power, and latency constraints [15].

This work addresses these challenges by investigating whether competitive NLI performance can be achieved through architectures specifically designed for resource-constrained training and deployment scenarios.

1.3 Objectives

This research aims to develop and evaluate DMoEBERT, an efficient encoder-only transformer architecture that addresses the computational efficiency challenges in NLI through the following specific objectives:

- **Architecture Design:** Develop an efficient encoder-only transformer that integrates sparse Mixture-of-Experts layers with Dynamic Tanh activation, optimized for resource-constrained training environments.
- **Sparse Expert Integration:** Implement and optimize noisy top-k routing mechanisms [5] within MoE layers to achieve computational efficiency while maintaining model expressiveness.
- **Normalization-Free Training:** Integrate Dynamic Tanh activation [7] to eliminate batch normalization and layer normalization overhead, reducing computational complexity and memory requirements.
- **Performance Validation:** Demonstrate competitive NLI performance on standard benchmarks (SNLI, ANLI) while achieving significant reductions in training time, parameter count, and computational requirements.
- **Comprehensive Ablation Studies:** Systematically evaluate the contribution of each architectural component through controlled experiments.

- **Efficiency Analysis:** Provide detailed computational efficiency comparisons against established baselines in terms of FLOPs, memory usage, and training time.
- **Resource-Constrained Training:** Validate the approach using accessible hardware (single P100 GPU) to demonstrate practical feasibility for researchers with limited resources.

1.4 Scope

This research operates within carefully defined boundaries to ensure focused investigation and reproducible results:

Architectural Scope:

- **Model Architecture:** Focus on encoder-only transformer architectures [10], specifically adapted for classification tasks rather than generative applications.
- **Mixture-of-Experts:** Implementation of sparse MoE layers [5, 6] with noisy top-k routing, examining expert counts and routing strategies suitable for efficiency.
- **Dynamic Tanh Layer:** Integration of Dynamic Tanh [7] as a normalization-free alternative to traditional activation-normalization combinations.
- **Dataset Coverage:** Evaluation on established NLI benchmarks including SNLI [1] and ANLI [3] to enable direct comparison with existing approaches.
- **Baseline Comparisons:** Systematic comparison against parameter-efficient fine-tuning approaches, specifically LoRA-adapted BERT-base and RoBERTa-base models [11].
- **Hardware Constraints:** All experiments conducted using Kaggle’s P100 GPU environment, ensuring reproducibility and accessibility for researchers with similar resource limitations.

1.5 Unfamiliarity of the Problem

The novelty of this work lies in applying techniques typically associated with large-scale, resource-intensive models, such as Sparse MoE and Dynamic Tanh—to small-scale, resource-constrained environments. Unlike prior research that targets trillion-parameter scaling [6], this work demonstrates that these ideas can be effectively adapted to settings with limited computational power.

We examine how sparse mixture-of-experts perform with noisy top-k routing when paired with Dynamic Tanh in normalization-free blocks in an encoder-only model under heavy

resource constraints both during training and deployment. By training DMoEBERT entirely from scratch without reliance on pre-trained checkpoints, this work provides evidence that competitive performance can be achieved in realistic low-resource scenarios.

1.6 Applications

The DMoEBERT architecture addresses critical needs across multiple domains where computational efficiency is extremely important.

Resource-Constrained Research:

- **Academic Research Groups:** Enables NLP research in institutions with limited computational budgets, democratizing access to advanced model architectures [?].
- **Educational Applications:** Provides accessible models for teaching and learning NLP concepts without requiring expensive infrastructure.
- **Developing Regions:** Supports NLP research and application development in regions with limited access to high-end computational resources.

Edge and Mobile Deployment:

- **Real-time Inference:** Enables deployment of NLI capabilities in applications requiring low-latency response times, such as conversational AI and content moderation systems.
- **IoT Devices:** Supports natural language understanding in Internet-of-Things applications where computational resources are severely constrained.
- **Privacy-Preserving Applications:** Enables on-device NLI processing, addressing privacy concerns associated with cloud-based inference.

Industrial Applications:

- **Startup Environments:** Provides cost-effective NLI solutions for startups and small companies that cannot afford large-scale model training and deployment.
- **Sustainable AI:** Contributes to environmentally sustainable AI practices by reducing energy consumption in training and deployment.
- **Rapid Prototyping:** Enables quick experimentation and prototyping of NLI-based applications without significant computational investment.

Downstream NLP Applications:

- **Question Answering Systems:** Lightweight inference for reading comprehension and factual question answering applications .
- **Dialogue and Conversational AI:** Efficient reasoning capabilities for chatbots and virtual assistants operating under computational constraints.
- **Content Analysis:** Real-time text classification and content understanding for social media monitoring, news analysis, and document processing.

Through this research, our objective is to demonstrate that advanced NLP capabilities can be achieved without requiring massive infrastructure investments, potentially accelerating the adoption and democratization of natural language understanding technologies across diverse application domains.

1.7 Project Planning

The following Gantt chart in Fig. 1.1 shows the planning of the project.

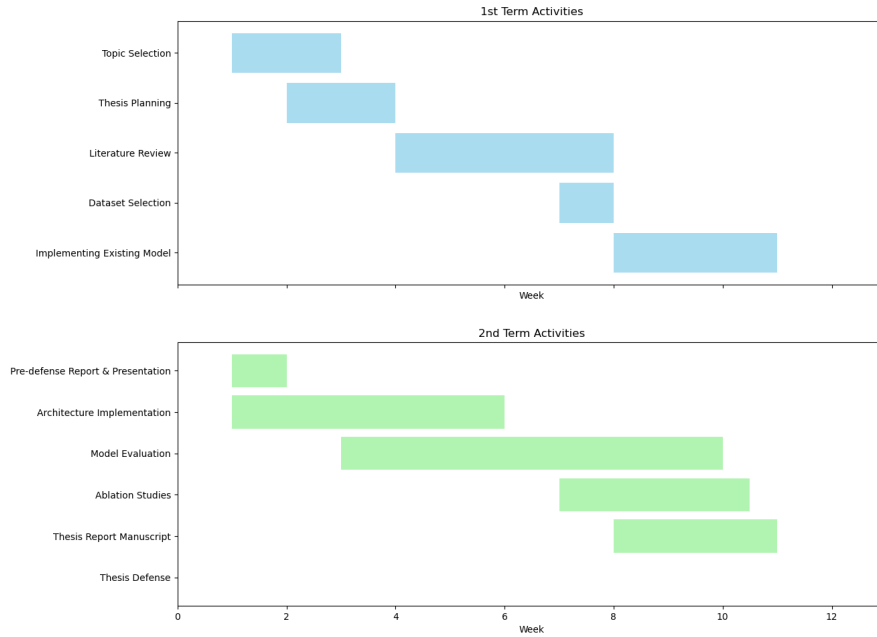


Figure 1.1: Gantt Chart of Project Planning

1.8 Organization of the report

The rest of the thesis is organized as follows:

Chapter I Provides an introduction to the background, problem statement, objectives, and organization of the thesis.

Chapter II Presents summaries of the related research works on llm architectures and natural language inference.

Chapter III Presents the methodology including model architecture, normalization strategy, and model configurations.

Chapter IV Details the implementation details, result and ablation studies.

Chapter V Details the project's Socio-Economic Impact and Sustainability.

Chapter VI Addresses the complex engineering problems associated with efficient natural language inference in resource-constrained Environments.

Chapter VII Discusses limitations, conclusions and future works.

CHAPTER II

Literature Review

2.1 Natural Language Inference (NLI) Datasets and Benchmarks

Natural Language Inference (NLI) has emerged as a fundamental task for evaluating natural language understanding capabilities, requiring models to determine logical relationships between premise-hypothesis pairs. The Stanford Natural Language Inference (SNLI) corpus [1], containing 570K sentence pairs collected through crowdsourcing, established the first large-scale benchmark for this task. However, SNLI’s limitation to simple sentences and specific domains motivated the development of MultiNLI [2], which expanded coverage to include diverse genres and writing styles across 433K examples, providing a more robust evaluation of model generalization.

The evolution continued with Adversarial NLI (ANLI) [3], which addresses a critical limitation of previous benchmarks: the potential for models to exploit spurious correlations rather than genuine reasoning. ANLI employs an iterative adversarial collection process where human annotators create examples specifically designed to challenge state-of-the-art models, resulting in progressively more difficult rounds (A1, A2, A3). This adversarial approach revealed that models achieving near-human performance on SNLI and MultiNLI still struggle significantly on more challenging examples, highlighting the gap between benchmark performance and true natural language understanding.

Despite their contributions, these benchmarks share common limitations that affect the broader NLI research landscape. The reliance on crowdsourced annotations introduces potential label noise and annotator bias, while the focus on English limits cross-linguistic evaluation. More importantly for our work, the computational requirements of achieving competitive performance on these benchmarks have created a resource barrier that limits accessibility to advanced NLI capabilities.

2.2 Transformer-based Architectures in NLI

The introduction of the Transformer architecture [10] fundamentally changed natural language processing by enabling highly parallelizable models with superior contextual representation capabilities. The self-attention mechanism allows models to capture long-range dependencies more effectively than previous recurrent architectures, while the parallel processing capability significantly reduces training time on appropriate hardware.

Building on this foundation, BERT [8] and RoBERTa [9] achieved state-of-the-art performance on NLI and related NLP tasks. However, these models typically contain hundreds of millions of parameters, making training and deployment computationally expensive. Their reliance on large-scale pre-training also poses challenges for research groups and practitioners with limited computational resources.

2.3 Mixture-of-Experts (MoE) Architectures

Mixture of Experts (MoE) architectures address scalability challenges through conditional computation, where different parts of the model are activated based on input characteristics. The foundational work by Jacobs et al. [4] introduced adaptive mixtures of local experts, using a gating network to learn input-dependent weightings of multiple expert networks. This approach enables models to increase capacity without proportional increases in computational cost, as only a subset of experts processes each input.

Shazeer et al. [5] advanced this concept significantly with Sparse MoE layers that replace traditional feed-forward networks in transformers. Their approach uses a learnable gating function to select the top-k experts for each input token, typically $k=1$ or $k=2$, achieving substantial parameter increases with minimal computational overhead. For example, their model achieved better performance than a dense baseline while using only 17% of the computational cost during inference.

The Switch Transformer [6] demonstrated the scalability potential of this approach, scaling to over 1.6 trillion parameters while maintaining efficiency through expert parallelism. By using $k=1$ routing and distributing experts across multiple devices, Switch Transformer achieved strong performance across diverse tasks while maintaining computational tractability. The model demonstrated that MoE architectures can achieve better scaling properties than dense models, with more favorable parameter-to-performance ratios.

Despite their efficiency benefits, these approaches have primarily been applied in high-resource environments, the interaction between MoE architectures and other efficiency techniques remains underexplored. Specifically, the combination of sparse expert selection with normalization-free training approaches has not been systematically investigated, despite the potential for excellent efficiency gains.

2.4 Parameter-Efficient Fine-Tuning: LoRA and Low-Rank Adaptations

Parameter-efficient fine-tuning has emerged as a promising approach to address the computational challenges of adapting large pre-trained models to specific tasks. Low-Rank Adaptation (LoRA) [11] represents a significant advancement in this area by introducing

the insight that the weight updates during fine-tuning lie in a low-dimensional subspace. This observation enables efficient adaptation through low-rank matrix decompositions rather than updating all model parameters.

LoRA works by decomposing weight updates into two low-rank matrices, $\Delta W = BA$, where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$. During fine-tuning, the original pre-trained weights remain frozen while only the low-rank adaptation matrices are updated. This approach reduces the number of trainable parameters dramatically—by factors of 10,000× in some cases—while maintaining performance comparable to full fine-tuning across various tasks including natural language inference.

The effectiveness of LoRA stems from its ability to capture the essential directions of weight updates without the overhead of full parameter modification. On NLI benchmarks, LoRA-adapted models achieve performance within 1-2% of fully fine-tuned counterparts while requiring significantly less memory and computational resources during training. The method’s efficiency makes it particularly attractive for researchers with limited computational budgets, as adaptation can be performed on consumer-grade hardware.

Moreover, LoRA’s effectiveness depends heavily on the quality and relevance of the pre-trained model to the target task. Even at deployment, the models retain all the parameters, but such huge number of parameters may not be necessary. This limitation suggests that developing efficient architectures tailored specifically for natural language inference could potentially achieve better efficiency-performance trade-offs than adapting general-purpose pre-trained models.

2.5 Normalization-free Transformers and Dynamic Tanh

Traditional transformers rely on normalization layers, such as LayerNorm, to stabilize training. However, normalization incurs computational overhead and introduces inefficiencies in distributed environments. Dynamic Tanh [7] has been proposed as an alternative, providing stability without normalization while mitigating vanishing or exploding gradients. Dynamic Tanh is particularly promising for resource-efficient models but as it is a very recent paradigm its combination with Sparse MoE layers has not been completely explored.

2.6 Research Gap

While large-scale models and MoE architectures have demonstrated remarkable performance, their reliance on massive computational resources makes them unsuitable for resource-constrained environments. Similarly, Dynamic Tanh has shown potential to remove normalization overhead, but its integration with MoE in practical low-resource training has not been fully studied. This gap motivates the development of DMoEBERT, an

encoder-only model that combines Sparse MoE with noisy top-k routing and Dynamic Tanh to achieve competitive NLI performance under limited computational budgets and make advanced natural language understanding capabilities accessible to a broader research community.

Table 2.1: Summary of Reviewed Literatures

Reference	Method	Remark
[10]	Transformer	Introduced the self-attention mechanism, enabling parallel training and effective modeling of long-range dependencies.
[8]	BERT	Utilizes bidirectional self-attention, capturing information from both directions in a sentence, excellent at contextual understanding.
[9]	RoBERTa	Optimized BERT variant trained longer on more data; improved performance at the cost of higher resource requirements.
[4]	Mixture of Experts (MoE)	A modular architecture that combines multiple specialized neural networks (“experts”) and dynamically routes each token through a learned gating mechanism, enabling efficient handling of complex tasks.
[5]	Sparse MoE	Replaces dense feed-forward layers with sparse experts; activates only top- k experts per input, significantly reducing inference cost.
[6]	Switch Transformer	Scaled MoE model to trillions of parameters using efficient expert routing ($k = 1$), demonstrating improved parameter-to-performance scaling.
[11]	LoRA	Parameter-efficient fine-tuning approach using low-rank decomposition; reduces trainable parameters by several orders of magnitude while maintaining accuracy.
[7]	Dynamic Tanh	Normalization-free activation function that stabilizes training and improves efficiency, avoiding LayerNorm overhead.

CHAPTER III

Methodology

3.1 DMoEBERT Architecture

DMoEBERT uses multi-head attention and sparse mixture-of-experts with noisy top-k router for high efficiency. We used Dynamic Tanh to implement normalization free blocks that further speeds up training and inference. We add learned positional embeddings to the input embeddings to encode the sequence position.

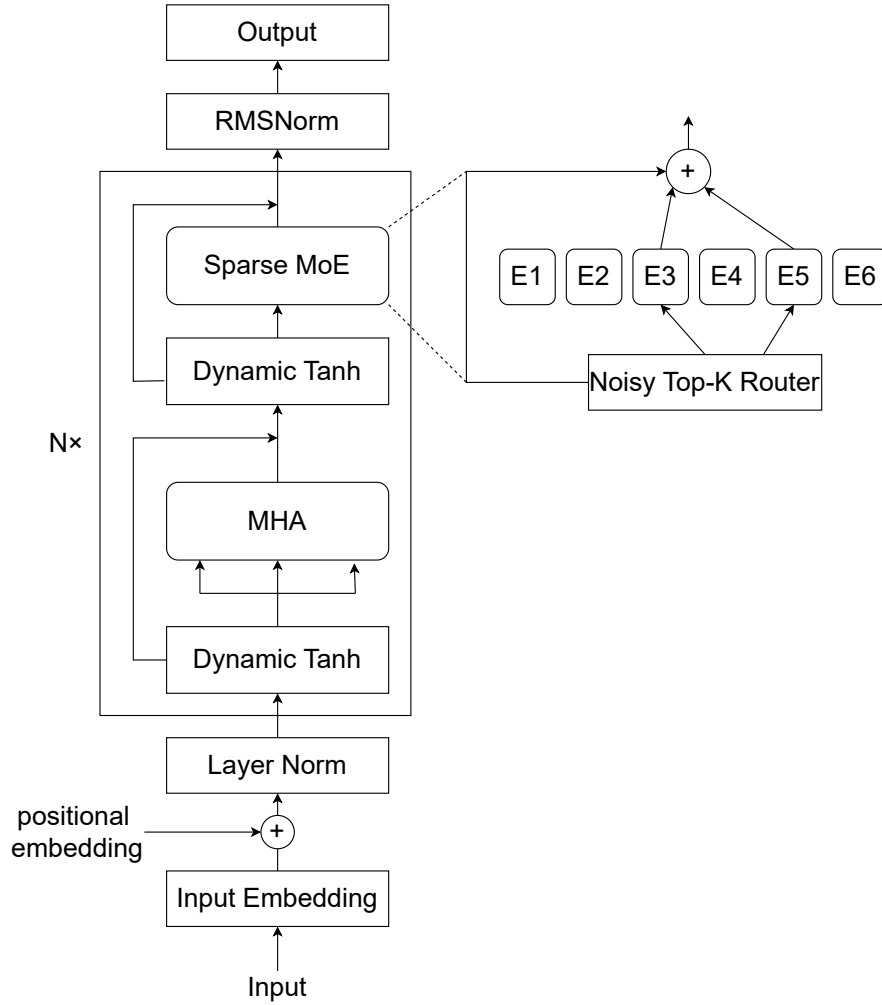


Figure 3.1: DMoEBERT Architecture

3.1.1 Multi-Head Attention (MHA)

Multi-Head Attention is a core component of transformer architectures [10] that allows the model to focus on different parts of the input sequence simultaneously. By computing multiple attention distributions (heads) in parallel, the model can capture diverse contextual relationships and dependencies.

Given input representations Q (queries), K (keys), and V (values), the scaled dot-product attention for a single head is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

where d_k is the dimensionality of the key vectors.

For multi-head attention, the outputs of all heads are concatenated and projected to the original embedding dimension:

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

where each $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$, and W_i^Q, W_i^K, W_i^V, W^O are learned projection matrices.

3.1.2 Dynamic Tanh

Normalization is critical to prevent vanishing or exploding gradients. It improves training speed, stability, and performance, and helps models converge faster. Dynamic Tanh [7] achieves the effects of normalization without the need to compute activation statistics.

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\text{Dynamnic Tanh}(x) = \gamma \cdot \tanh(\alpha \cdot x) + \beta$$

3.1.3 Sparse MoE Integration

Expert Networks Each expert is a simple feed-forward neural network. We expand the hidden dimension to $2 \times d_{\text{embed}}$ in the intermediate layer, before projecting it back to the original embedding dimension.

$$\text{Expert}(x) = \text{Dropout}(\text{Linear}(\text{GELU}(\text{Linear}(x))))$$

Noisy Top-K Router The router determines which experts process each token. To prevent deterministic routing patterns, we add learnable noise with the logits. The routing process

is defined as:

$$\begin{aligned}\text{logits} &= W_r \times x, \quad \text{noise} = W_n \times x \\ \text{noisy_logits} &= \text{logits} + \epsilon \times \text{softplus}(\text{noise})\end{aligned}$$

where $\epsilon \sim \mathcal{N}(0, 1)$ is Gaussian noise, and $W_r, W_n \in \mathbf{R}^{d_{\text{embed}} \times N_{\text{experts}}}$ are learnable parameters. The top- k experts are selected based on noisy logits, while the remaining experts are masked with $-\infty$ values. This sparse selection mechanism significantly reduces computational cost and inference speed.

Expert Capacity and Load Balancing We implement capacity constraints and an auxiliary loss function to prevent expert collapse and ensure balanced utilization.

$$\text{tokens_per_expert} = \frac{T \times k \times \text{capacity_factor}}{\text{num_experts}}$$

$$\mathcal{L}_{\text{balance}} = \left(\sum_{i=1}^{N_{\text{experts}}} f_i \cdot p_i \right) \times \text{num_experts}$$

where T is the number of total tokens, k is the number of selected experts, f_i is the fraction of tokens routed to expert i , and p_i is the mean routing probability for expert i . This maintains uniform distribution of tokens among experts.

3.1.4 Normalization Strategy

We use Layer Normalization after input embeddings and RMSNorm before the final output. Our experiments show that normalizing the input embeddings before passing them to the blocks increases performance.

3.2 Model Configurations

Table 3.1 shows both model configurations for both DMoEBERT-23M and DMoEBERT-45M

Table 3.1: Model configuration specifications

Configuration	DMoEBERT-23M	DMoEBERT-45M
d_{embed}	256	384
Attention Heads	4	6
Experts	6	6
Layers	8	8
Capacity Factor	1.15	1.15

Both models use a sequence length of 128 tokens and vocabulary size of 30,522 (BERT tokenizer).

CHAPTER IV

Implementation and Results

4.1 Implementation Details

4.1.1 Dataset

We trained on the concatenated SNLI [1] and MNLI [2] training sets (942K samples). Validation uses SNLI+MNLI matched (19.6K samples) and MNLI mismatched (9.8K samples). We evaluated in the SNLI test set and performed a zero-shot evaluation in ANLI R2 / R3 [3] to assess adversarial robustness.

For ablation studies, we include ANLI R1/R2/R3 in the training data (1.1M total samples) with combined validation sets (22.8K samples). The final evaluation remains in the SNLI test set and the ANLI R2/R3 test sets.

4.1.2 Baseline Comparison

We compared our models against BERT-base [8] (110M parameters), RoBERTa-base [9] (125M parameters). We finetuned these models on our dataset using parameter-efficient fine-tuning (PEFT) with LoRA adapters [11].

4.1.3 Training Strategy

For training, the model weights are initialized using a normal distribution $\sim \mathcal{N}(0, 0.02)$. For regularization, we set gradient clipping to 1.0 and dropout to 0.1. We used mixed precision training using bfloat16 format for 4 epochs. We set batch sizes to 512 and 256, maximum learning rate $6e-4$, $4.5e-4$ and minimum learning rate $6e-5$, $4.5e-5$ for DMoEBERT-23M and DMoEBERT-45M respectively.

For finetuning BERT-base and RoBERTa-base, we set batch sizes to 64 and 128 respectively, LoRA rank to 8 and α to 16, maximum learning rate $2e-4$, minimum learning rate $2e-5$ and finetune for 1 epoch. For the learning rate scheduler, we used cosine decay with 3% warmup and 25% plateau.

$$\eta(t) = \begin{cases} \frac{t+1}{\text{warmup_steps}} \cdot \eta_{\max}, & 0 \leq t < \text{warmup_steps}, \\ \eta_{\max}, & \text{warmup_steps} \leq t < \text{plateau}, \\ \eta_{\min} + \frac{1}{2} \left(1 + \cos \left(\pi \cdot \frac{t - \text{plateau}}{\text{max_steps} - \text{plateau}} \right) \right) (\eta_{\max} - \eta_{\min}), & \text{plateau} \leq t \leq \text{max_steps}, \\ \eta_{\min}, & t > \text{max_steps}. \end{cases}$$

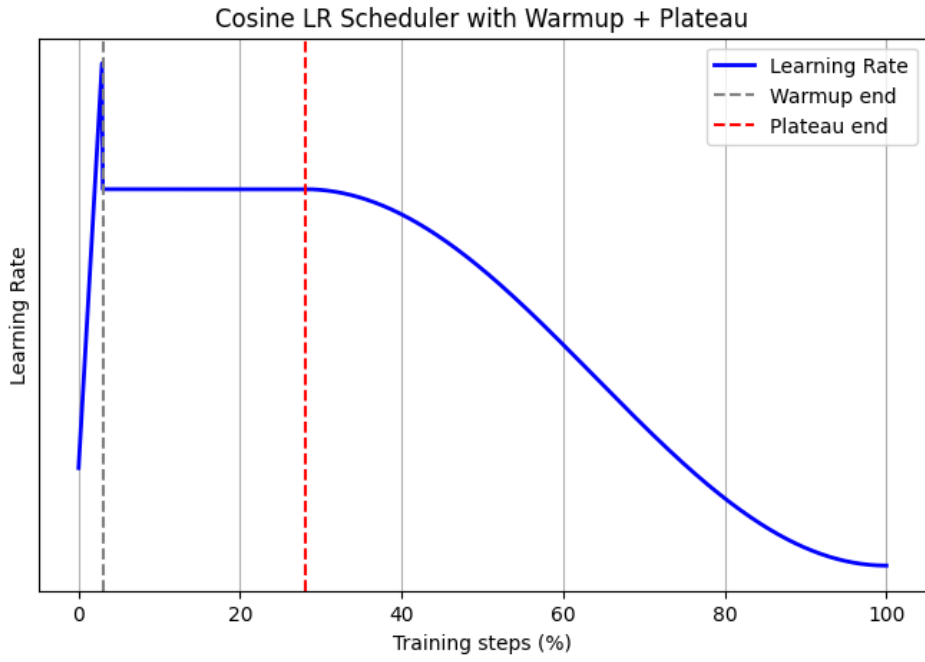


Figure 4.1: Learning Rate Scheduler Curve

For assessing the adversarial robustness of the models first we do zero-shot classification on ANLI R2/R3, then we augment the dataset with ANLI R1/R2/R3 and again train both models and perform the classification again.

We conducted all our experiments on Kaggle's P100 GPU (16GB memory).

4.2 Result Analysis

4.2.1 Performance overview

DMoEBERT-23M achieves 82.7% accuracy and DMoEBERT-45M achieves 83.7% accuracy on the Stanford Natural Language Inference (SNLI) dataset, while the fine-tuned BERT-base and RoBERTa-base models achieve 87.4% and 88.8% accuracy on SNLI. Although the accuracy is 4-5% below fine-tuned baselines, but DMoEBERT-23M and DMoEBERT-45M uses significantly fewer parameters. For each token DMoEBERT-23M activates 63% and DMoEBERT-45M activates 58% of the parameters which trims down the computational requirements even further, resulting in high efficiency. Our models require significantly fewer FLOPs (up to 13.5× less) and demonstrate strong scalability. Notably, DMoEBERT-23M took even less time to train than fine-tuning both baselines. To ensure reproducibility and statistical robustness, we conducted each experiment multiple times using different random seeds and hyperparameter configurations. The reported results demonstrate consistent accuracy across runs, with standard deviations of 0.33% for DMoEBERT-23M and 0.31% for DMoEBERT-45M over three representative seeds (42, 123, 1337).

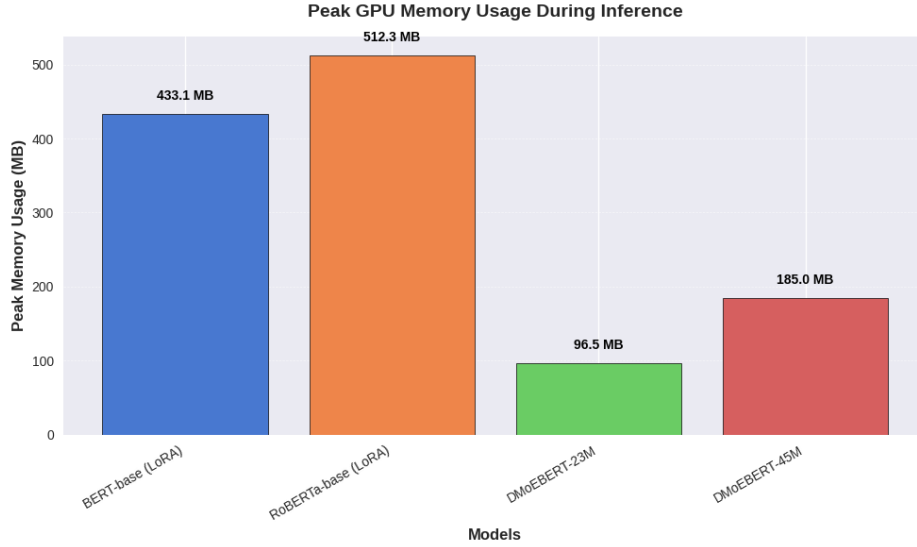


Figure 4.2: Peak Memory Usage by Different Models

Detailed findings are shown in Table 4.1, 4.2, and 4.3.

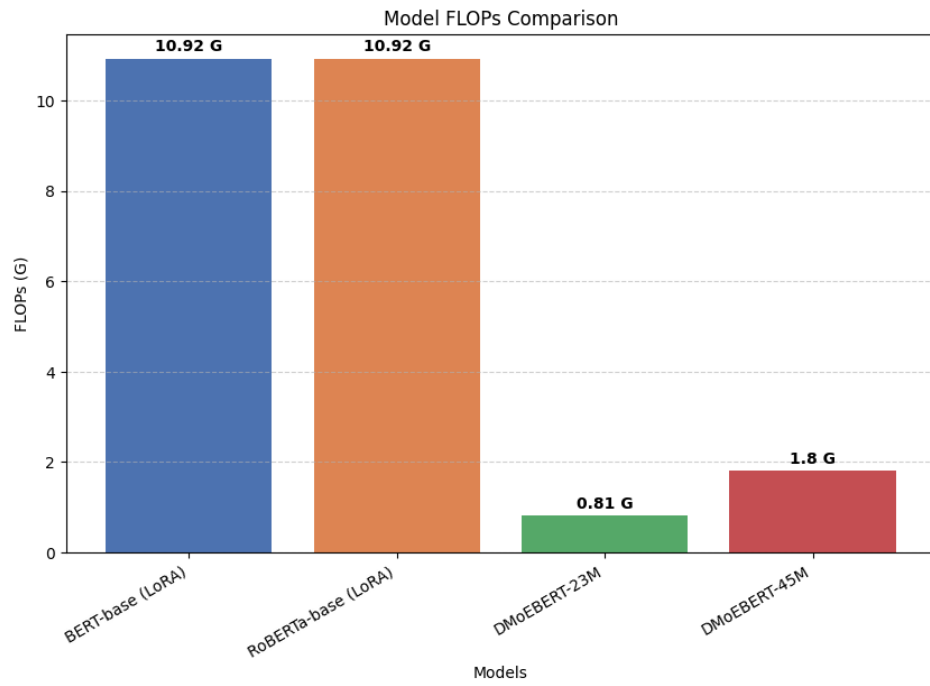


Figure 4.3: FLOPs required by Different Models

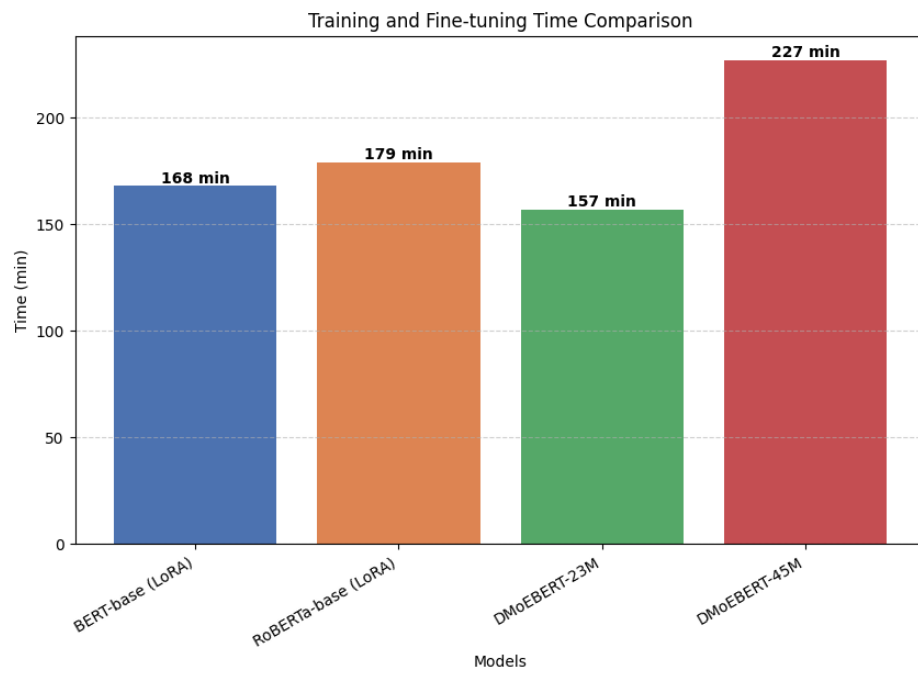


Figure 4.4: Time required to fine-tune BERT-base and RoBERTa-base and train DMoEBERT-23M and DMoEBERT-45M

Table 4.1: Performance comparison and ablation study on SNLI and ANLI test sets.
Acc.=Accuracy, F1-M=F1-macro

Model	SNLI(%)		ANLI-R2(%)		ANLI-R3(%)		Training
	Acc.	F1-M	Acc.	F1-M	Acc.	F1-M	Time
BERT-base (LoRA)	87.4	87.3	28.7	28.1	31.2	30.9	2h 48m
RoBERTa-base (LoRA)	88.8	88.8	32.8	21.9	32.7	21.1	2h 59m
DMoEBERT-23M	82.7	82.6	27.7	27.6	30.7	30.6	2h 37m
DMoEBERT-45M	83.7	83.7	28.8	28.7	28.4	28.4	3h 43m
<i>Ablation Study</i>							
DMoEBERT-23M w/o LayerNorm	78.1	78.0	28.3	27.9	29.7	29.4	2h 37m
DMoEBERT-45M w/o LayerNorm	79.6	79.4	28.5	28.1	28.8	28.6	3h 43m
DMoEBERT-23M (+ANLI)	83.0	82.9	37.2	36.9	37.6	37.1	3h 14m
DMoEBERT-45M (+ANLI)	83.6	83.6	40.3	40.1	38.7	38.4	4h 54m

4.2.2 Ablation Studies

LayerNorm: We trained both DMoEBERT-23M and DMoEBERT-45M without normalizing the input embeddings from scratch, and observed an accuracy drop of over 4%. Normalization also ensured smoother gradient flow and kept training more stable.

Adding ANLI into the train set: We augmented our training dataset (SNLI+MNLI) with ANLI R1/R2/R3, and trained both models from scratch on this extended dataset.

Our detailed findings are shown in Table 4.1.

Table 4.2: Per-class performance on SNLI test set. Ent. = Entailment, Neu. = Neutral, Con. = Contradiction.

Model	Precision			Recall			F1-Score		
	Ent.	Neu.	Con.	Ent.	Neu.	Con.	Ent.	Neu.	Con.
BERT-base (LoRA)	0.89	0.85	0.88	0.88	0.83	0.90	0.89	0.84	0.89
RoBERTa-base (LoRA)	0.90	0.85	0.91	0.90	0.86	0.91	0.90	0.86	0.91
DMoEBERT-23M	0.86	0.78	0.84	0.87	0.77	0.84	0.86	0.77	0.84
DMoEBERT-45M	0.87	0.79	0.85	0.87	0.78	0.86	0.87	0.79	0.86
DMoEBERT-23M w/o LN	0.80	0.74	0.80	0.86	0.71	0.78	0.83	0.72	0.79
DMoEBERT-23M (+ANLI)	0.87	0.77	0.85	0.86	0.78	0.85	0.87	0.78	0.85
DMoEBERT-45M w/o LN	0.82	0.75	0.82	0.86	0.73	0.79	0.84	0.74	0.80
DMoEBERT-45M (+ANLI)	0.87	0.78	0.86	0.86	0.80	0.85	0.86	0.79	0.86

Table 4.3: Computational efficiency comparison

Model	Params (M)	FLOPs (G)	Size (MB)	Peak Memory Usage (MB)
BERT-base (LoRA)	109.8	10.92	419.93	433.13
RoBERTa-base (LoRA)	125.5	10.92	482.28	512.31
DMoEBERT-23M	22.6	0.81	86.39	96.45
DMoEBERT-45M	44.9	1.80	171.51	184.98

4.2.3 Adversarial Robustness Analysis

Our models show a trade-off between efficiency and robustness that is characteristic of resource-constrained architectures on ANLI datasets (Table ??), which we attribute to two key factors: (1) **from-scratch training brittleness**, where models trained on limited data develop domain-specific decision boundaries, and (2) **sparse expert activation**, where adversarial examples may route to undertrained experts in the MoE layers.

Adding ANLI to the training data results in significant improvements (DMoEBERT-45M: +11.5% on ANLI-R2) with little effect on SNLI performance (Table 3), demonstrating the models’ ability for multi-domain learning. However, performance is still below the results reported in prior work, which is in line with research showing that smaller models are less robust because of their lower representational capacity [12]

Detailed findings are shown in Table 4.4, 4.5.

Table 4.4: Per-class performance on ANLI R2 test set. Ent. = Entailment, Neu. = Neutral, Con. = Contradiction.

Model	Precision			Recall			F1-Score		
	Ent.	Neu.	Con.	Ent.	Neu.	Con.	Ent.	Neu.	Con.
BERT-base (0-shot)	0.23	0.27	0.33	0.17	0.28	0.41	0.19	0.27	0.37
RoBERTa-base (0-shot)	0.24	0.39	0.33	0.03	0.07	0.88	0.06	0.12	0.48
DMoEBERT-23M (0-shot)	0.27	0.26	0.30	0.33	0.25	0.25	0.30	0.26	0.27
DMoEBERT-45M (0-shot)	0.30	0.27	0.30	0.34	0.25	0.27	0.32	0.26	0.28
DMoEBERT-23M (+ANLI)	0.36	0.40	0.36	0.43	0.41	0.27	0.39	0.40	0.31
DMoEBERT-45M (+ANLI)	0.39	0.42	0.41	0.46	0.42	0.32	0.42	0.42	0.36

Table 4.5: Per-class performance on ANLI R3 test set. Ent. = Entailment, Neu. = Neutral, Con. = Contradiction.

Model	Precision			Recall			F1-Score		
	Ent.	Neu.	Con.	Ent.	Neu.	Con.	Ent.	Neu.	Con.
BERT-base (0-shot)	0.32	0.31	0.30	0.25	0.37	0.32	0.28	0.34	0.31
RoBERTa-base (0-shot)	0.28	0.33	0.33	0.03	0.06	0.90	0.05	0.10	0.48
DMoEBERT-23M (0-shot)	0.31	0.32	0.29	0.34	0.32	0.26	0.32	0.32	0.27
DMoEBERT-45M (0-shot)	0.29	0.29	0.27	0.28	0.30	0.27	0.29	0.30	0.27
DMoEBERT-23M (+ANLI)	0.38	0.39	0.35	0.40	0.47	0.27	0.39	0.42	0.30
DMoEBERT-45M (+ANLI)	0.39	0.39	0.37	0.41	0.45	0.30	0.40	0.42	0.33

4.2.4 Confusion Matrix Analysis

The confusion matrices reveal that both models struggle the most with the “Neutral” class, which is consistent with previous NLI research. Our models show similar error patterns to fine-tuned models, indicating that the architectural changes preserve the fundamental reasoning capabilities while improving efficiency.

We performed zero-shot classification on ANLI-R2 and ANLI-R3 dataset. After augmenting the dataset with ANLI data we see a significant increase in models’ adversarial robustness.

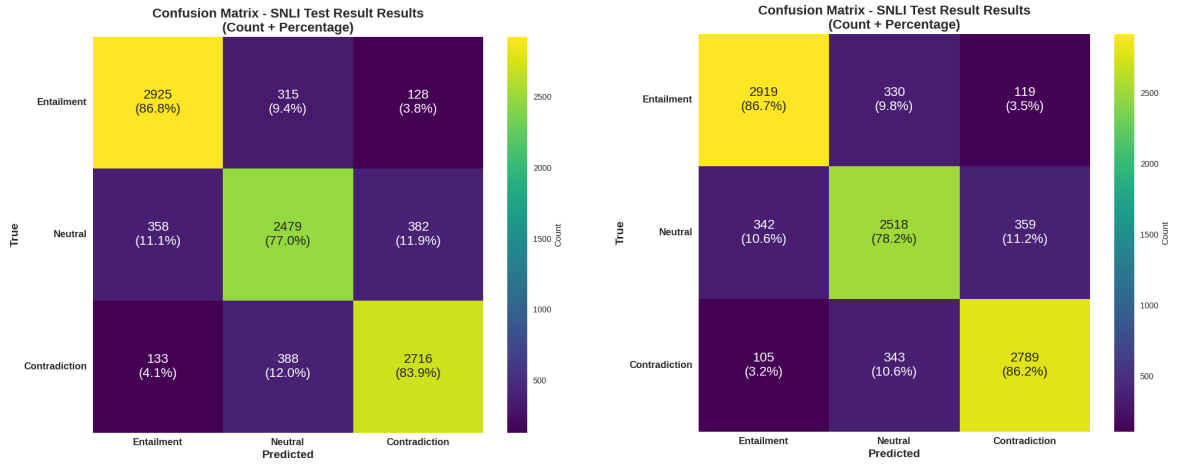


Figure 4.5: Confusion matrices for DMoEBERT-23M (left) and DMoEBERT-45M (right).

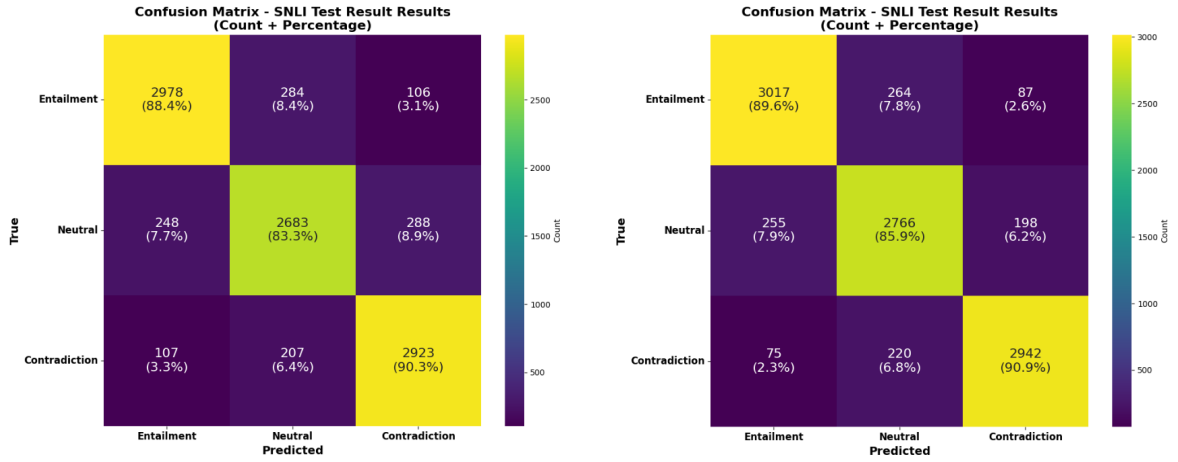


Figure 4.6: Confusion matrices for BERT (left) and RoBERTa (right).

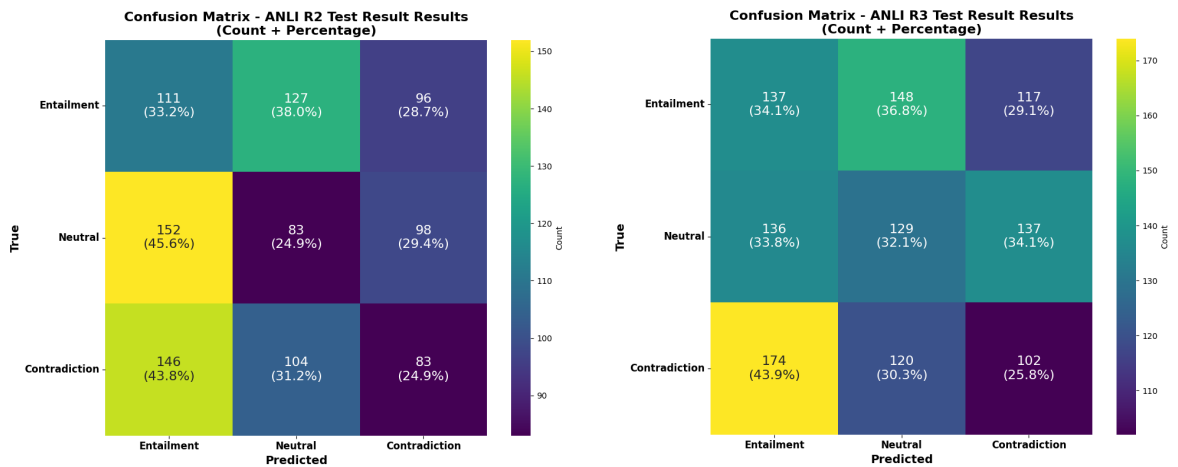


Figure 4.7: Confusion matrices for DMoEBERT-23M ANLI-R2 (left) and ANLI-R3 (right) (zero-shot).

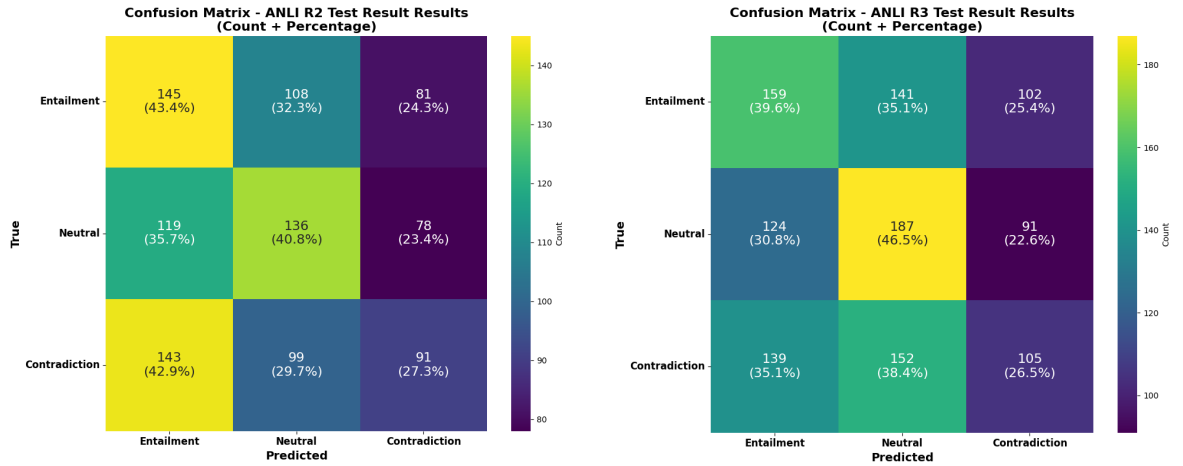


Figure 4.8: Confusion matrices for DMoEBERT-23M ANLI-R2 (left) and ANLI-R3 (right) (ANLI added)

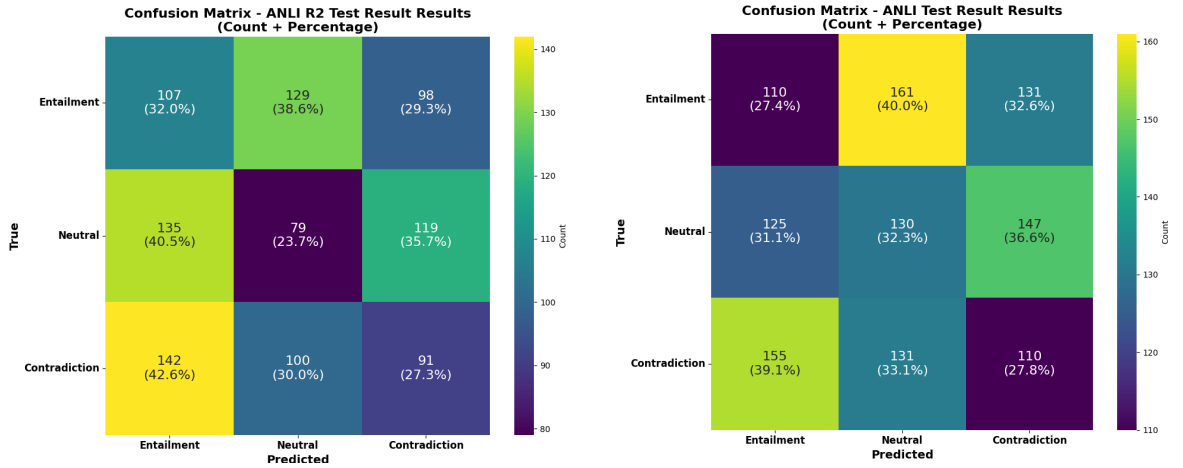


Figure 4.9: Confusion matrices for DMoEBERT-45M ANLI-R2 (left) and ANLI-R3 (right) (zero-shot).

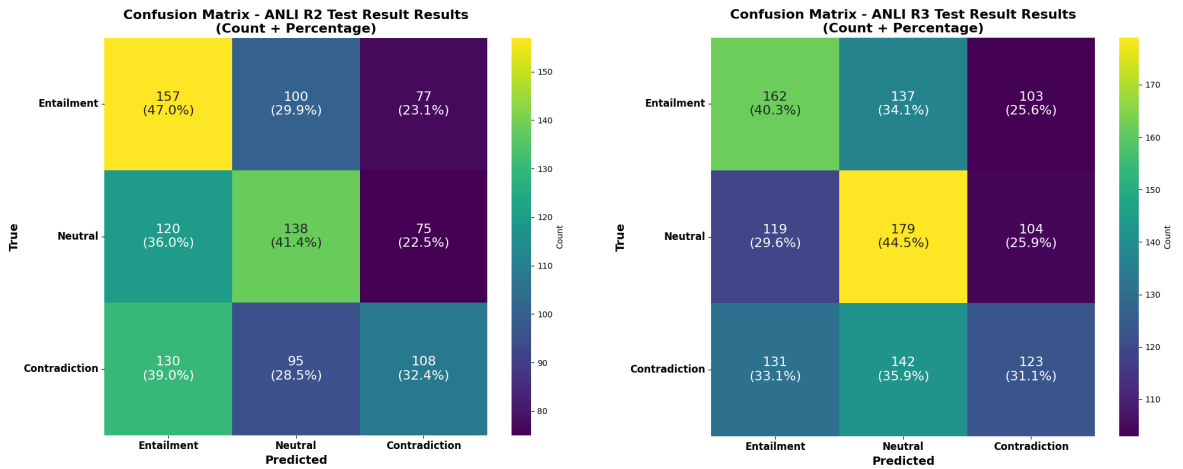


Figure 4.10: Confusion matrices for DMoEBERT-45M ANLI-R2 (left) and ANLI-R3 (right) (ANLI added).

4.2.5 Training Dynamics

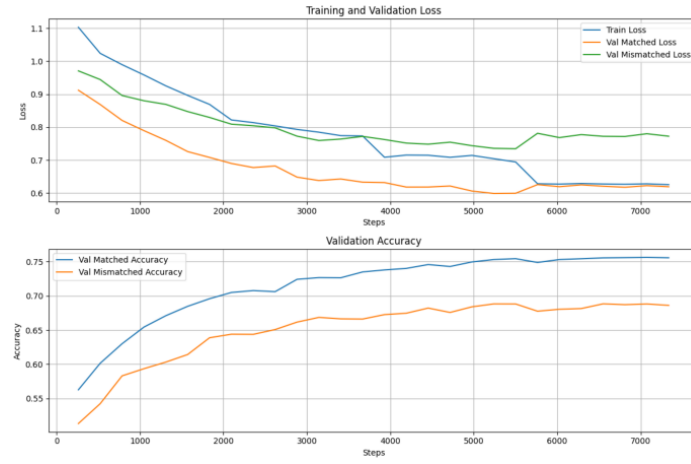


Figure 4.11: DMoEBERT-23M Training loss curves (top) and validation accuracy (bottom).

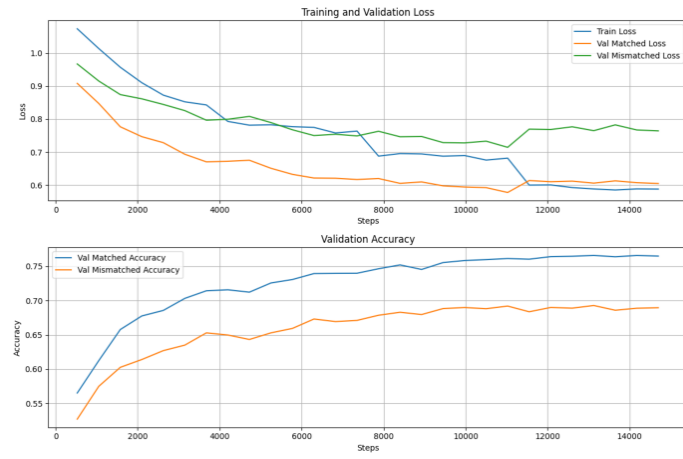


Figure 4.12: DMoEBERT-45M Training loss curves (top) and validation accuracy (bottom).

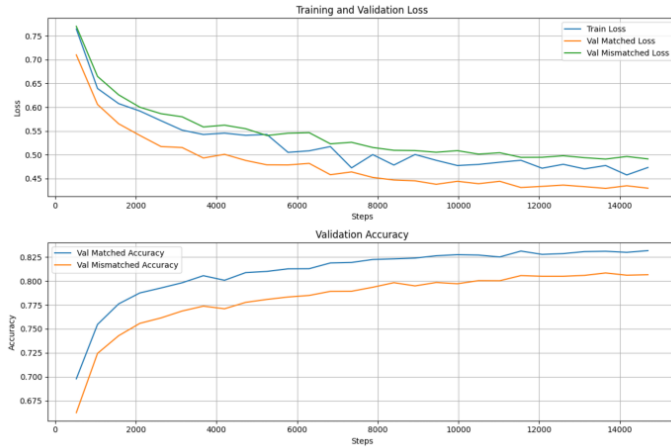


Figure 4.13: BERT-base Training loss curves (top) and validation accuracy (bottom).

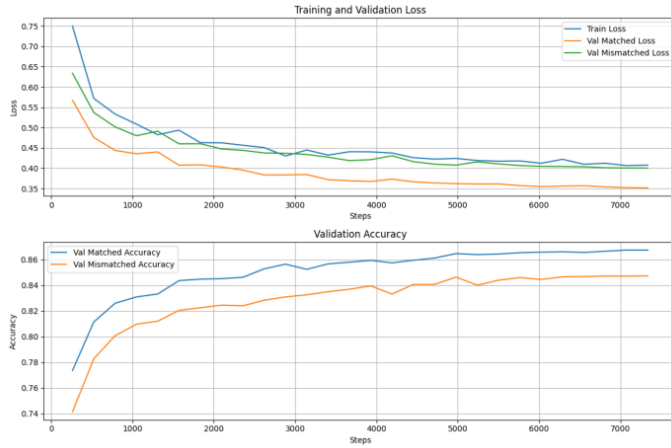


Figure 4.14: RoBERTa-base Training loss curves (top) and validation accuracy (bottom).

4.3 Objectives Achieved

The following objectives outlined at the beginning of the work have been successfully achieved:

- **Architecture Design:** Designed and implemented DMoEBERT, an efficient encoder-only transformer integrating sparse Mixture-of-Experts layers with Dynamic Tanh activation, optimized for resource-constrained training environments.
- **Sparse Expert Integration:** Successfully implemented and optimized noisy top-k routing mechanisms [5] within MoE layers, achieving computational efficiency while maintaining model expressiveness.
- **Normalization-Free Training:** Integrated Dynamic Tanh activation [7], resulting in normalization-free blocks. We eliminated batch normalization and layer normalization overhead, reducing computational complexity and memory usage.

- **Performance Validation:** Demonstrated competitive Natural Language Inference performance on standard benchmarks (SNLI, ANLI) with notable reductions in training time, parameter count, and FLOPs.
- **Comprehensive Ablation Studies:** Conducted systematic ablation studies against BERT-base and RoBERTa-base to evaluate our architectural design choices.
- **Efficiency Analysis:** Provided detailed comparisons against strong baselines in terms of FLOPs, peak memory usage, and training time, model size, validating efficiency claims.
- **Resource-Constrained Training:** Validated the proposed approach on accessible hardware (single P100 GPU), demonstrating its practical feasibility for researchers with limited computational resources.

4.4 Financial Analysis and Budget

As all experiments were conducted using Kaggle’s freely available P100 GPUs, no financial costs were incurred for computation or infrastructure.

CHAPTER V

Socio-Economic Impact and Sustainability

5.1 Societal, Health, Safety, Legal, and Cultural Impact

The development of DMoEBERT, a resource-efficient NLI model, has several positive societal implications. By enabling high-quality natural language understanding with limited computational resources, this project democratizes access to AI technologies, allowing smaller research labs and organizations in developing regions to benefit from state-of-the-art NLP models.

From a health and safety perspective, NLI models can assist in automated monitoring of medical texts, patient records, and health advisories, potentially improving decision-making in healthcare systems. In addition, ensuring ethical usage of such models mitigates risks related to misinformation and biased decision-making.

Legally, models trained on publicly available datasets adhere to existing copyright and data usage regulations, reducing risks of infringement. Culturally, DMoEBERT can support multilingual or domain-specific language tasks, promoting inclusivity and bridging communication gaps across diverse communities.

5.2 Environmental Impact and Sustainability

The project emphasizes computational efficiency, which directly reduces energy consumption during model training and inference. By training models from scratch on a single Kaggle P100 GPU instead of large-scale clusters, the carbon footprint is significantly lower compared to conventional large-scale NLP models.

The design choices, including Sparse Mixture-of-Experts and Dynamic Tanh layers, further optimize resource usage without compromising performance. These considerations contribute to sustainable AI development practices, aligning the project with broader environmental and energy conservation goals.

Overall, DMoEBERT demonstrates that high-performing NLP models can be both ****societally beneficial and environmentally sustainable****, enabling responsible deployment in real-world applications.

CHAPTER VI

Addressing Complex Engineering Problems Associated with Efficient Natural Language Inference in Resource-Constrained Environments

6.1 Domain Overview

Developing efficient natural language inference models for resource-constrained environments represents one of the most challenging problems in modern natural language processing. While large-scale transformer models have achieved remarkable performance on NLI tasks, their computational requirements create significant barriers for deployment in practical scenarios. This chapter discusses the complex engineering problems associated with developing DMoEBERT, a sparse mixture-of-experts encoder architecture that addresses these fundamental challenges through innovative architectural design and training methodologies.

6.2 Complex Engineering Problems Associated with Efficient NLI

The development of DMoEBERT for efficient natural language inference presents a multifaceted engineering challenge that encompasses algorithmic innovation, resource optimization, and performance validation. Table 6.1 characterizes the complexity of this engineering problem.

6.3 Complex Engineering Activities Associated with Efficient NLI

The research and development process for DMoEBERT involved numerous complex engineering activities that required careful coordination and innovative problem-solving approaches. Table 6.2 outlines the key activities and their associated complexities.

Table 6.1: Complex Engineering Problems Associated with Efficient NLI

Attributes	Complex Engineering Problem Associated with Efficient NLI
Depth of Knowledge Required (P1)	The thesis requires comprehensive understanding of multiple advanced concepts including transformer architectures, attention mechanisms, moe routing algorithms, sparse activation techniques, normalization-free training methodologies, and their mathematical foundations. Deep knowledge of NLI benchmarks, evaluation metrics, and computational efficiency analysis was essential for developing a viable solution.
Range of Conflicting Requirements (P2)	The primary challenge involved balancing multiple conflicting objectives: achieving competitive accuracy while minimizing computational resources, maintaining model expressiveness with sparse expert activation, training from scratch without pre-trained weights while competing against fine-tuned large models, and optimizing for both training efficiency and inference speed simultaneously.
Depth of Analysis Required (P3)	Extensive analysis was required including architectural component ablation studies, hyperparameter optimization for sparse routing mechanisms, computational efficiency profiling (FLOPs, memory usage, training time), adversarial robustness evaluation, and comparative performance analysis against established baselines.
Familiarity of Issues (P4)	The integration of Dynamic Tanh with sparse mixture-of-experts in encoder-only architectures for NLI represents a novel combination not extensively explored in existing literature. The challenge of achieving competitive performance with significantly fewer parameters while training from scratch in resource-constrained environments introduced unfamiliar optimization challenges and architectural considerations.
Extent of Applicable Codes (P5)	The research involved implementing complex algorithms including noisy top-k routing mechanisms, expert load balancing, normalization-free transformer blocks, and custom training procedures. All implementations followed software engineering best practices with proper attribution to referenced algorithms and adherence to reproducibility standards. The codebase required integration of multiple state-of-the-art techniques adapted for resource-constrained scenarios.
Extent of Stakeholder Involvement and Conflicting Requirements (P6)	The thesis addresses needs of multiple stakeholder groups with potentially conflicting requirements: academic researchers requiring accessible model architectures, industry practitioners needing deployable solutions, and the broader AI community concerned with sustainable computing practices. Balancing performance expectations with resource constraints while maintaining research rigor created complex requirement negotiations.
Interdependence (P7)	The architectural components exhibit high interdependence where sparse expert selection affects training dynamics, normalization strategies impact convergence behavior, routing mechanisms influence computational efficiency, and dataset augmentation strategies affect adversarial robustness. Changes to any component required careful consideration of system-wide effects and potential performance implications across all evaluation metrics.

Table 6.2: Complex Engineering Activities Associated with Efficient NLI

Attributes	Complex Engineering Activities Associated with Efficient NLI
Resources (A1)	The thesis required creative problem-solving to overcome significant resource constraints, utilizing Kaggle’s P100 GPU environment effectively through mixed precision training, gradient accumulation strategies, and memory optimization techniques. Extensive literature review spanning transformer architectures, mixture-of-experts methodologies, and efficiency optimization guided the innovative combination of techniques. Supervisor guidance and access to computational resources through institutional partnerships enabled successful completion of comprehensive experiments.
Level of Interaction (A2)	The research involved interaction with multiple communities including the natural language processing research community through literature review and baseline comparisons, the mixture-of-experts research community for architectural insights, and the sustainable AI community regarding efficiency considerations. Regular meetings with supervisors, collaboration with department faculty, and engagement with peer researchers provided essential feedback for refining the approach and validating experimental results.
Innovation (A3)	The thesis demonstrated innovation through the novel integration of Dynamic Tanh normalization-free training with sparse mixture-of-experts routing in encoder-only architectures. The development of DMoEBERT represents a creative application of efficiency techniques typically reserved for large-scale models to resource-constrained scenarios. The innovative training methodology achieving competitive performance from scratch without pre-trained weights challenges conventional wisdom about the necessity of large-scale pre-training for effective NLI models.
Consequences for Society and Environment (A4)	The research directly addresses the environmental impact of large-scale AI model training and deployment by demonstrating that effective natural language understanding can be achieved with significantly reduced computational requirements. The work contributes to democratizing access to advanced NLP capabilities for researchers and organizations with limited resources, potentially accelerating innovation in underserved communities. The emphasis on sustainable AI practices aligns with growing concerns about the environmental footprint of large language models.
Familiarity (A5)	The thesis explored relatively uncharted territory by applying mixture-of-experts architectures to small-scale, resource-constrained natural language inference tasks. The combination of normalization-free training with sparse expert routing in encoder-only models required developing new insights about training dynamics and optimization strategies. The project demonstrated adaptability in addressing unfamiliar challenges through principled experimentation and systematic evaluation, contributing novel methodologies to the field of efficient natural language processing.

CHAPTER VII

Discussion & Conclusions

7.1 Discussion

Our findings highlight several important aspects of DMoEBERT’s behavior. First, we observe a clear *performance–efficiency trade-off*: while DMoEBERT does not fully match the accuracy of BERT-base, its ability to be trained from scratch under resource-constrained settings makes it a practical alternative for low-resource environments.

Second, the results confirm the persistent difficulty of the *Neutral* class, consistent with prior work in natural language inference. This indicates that the challenge is not unique to our architecture, but rather reflects a broader limitation in current approaches to modeling ambiguous semantic relationships. Future work could therefore explore specialized mechanisms for improved neutral-class handling.

Finally, the encouraging performance of smaller DMoEBERT variants points toward *scalability implications*. These results suggest that even more lightweight configurations could retain acceptable accuracy, thereby broadening opportunities for deployment in scenarios with strict computational or energy constraints.

7.2 Limitations

Limited Evaluation Scope We focused mainly on the SNLI evaluation. Testing across multiple NLI datasets and other NLP tasks would demonstrate broader applicability of our approach.

Comparison Fairness For the baseline comparison, we fine-tuned BERT-base [8] and RoBERTa-base [9] on our dataset, but training them from scratch under identical conditions would provide a fairer comparison.

7.3 Conclusions

In this thesis, we introduce DMoEBERT, a sparse MoE encoder-only architecture. Our models achieved competitive results with 82.7% and 83.7% SNLI accuracy using only 23M / 45M parameters. Augmenting the training data with ANLI significantly improved adversarial robustness with gains of +9.5-11.5%, while maintaining in-domain

performance. The high efficiency and lightweight nature of the models make them suitable for use in resource-constrained environments.

7.4 Future Works

Future work includes making our models more adversarially robust, examining performance on additional datasets, and cross-domain transfer capabilities.

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